AKARSH NAVHULE

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PROFESSIONAL SUMMARY

Full Stack AI Engineer with 2 years building production AI systems — RAG pipelines, LLM agents, diffusion model workflows, and GCP-deployed applications. Experienced shipping AI products end-to-end from model integration to frontend.

TECHNICAL SKILLS

Languages	Python, JavaScript
Frontend	React.js, Next.js, Tailwind CSS
Backend	FastAPI, Node.js, REST API Design
AI / ML	RAG, vector search, embeddings, fine-tuning (LoRA), agentic workflows, multimodal — ChromaDB, LLM APIs (Gemini, OpenAI), ComfyUI, n8n
Databases	MongoDB, PostgreSQL, ChromaDB
Cloud & DevOps	GCP (Compute Engine, Cloud Storage), Nginx, Shell scripts

WORK EXPERIENCE

Full Stack Developer · **L&T Finance** — Bengaluru, India — Jul 2024 – Oct 2025

- Built a high-throughput FastAPI service using AWS Rekognition for facial-recognition attendance, handling 500+ concurrent API calls/second with async batching and multi-threading.
- Developed a React.js analytics dashboard surfacing real-time attendance states; integrated MongoDB for persistent event logging and audit trails.
- Used ChatGPT for context-aware development; estimated 2x faster delivery compared to prior sprint velocity.

Freelance Full Stack Developer · **Self-Employed** — Bengaluru, India — Nov 2025 – Mar 2026

- Built **Malnadrealty.com** ↗ — a NoBroker-style property marketplace; sole engineer for end-to-end GCP deployment (Compute Engine, Nginx, MongoDB, PostgreSQL), OTP login, WhatsApp notifications, and headless WordPress CMS across 150+ REST APIs.

PROJECTS

01 AI Digital Twin — Face, Voice & Video Synthesis ↗ GitHub ↗ Portfolio

ComfyUI · FLUX LoRA · Chatterbox (voice cloning) · Infinite Talk · Hugging Face · Jarvis Labs GPU · ngrok

- Fine-tuned a FLUX LoRA on 15 personal photos with custom trigger token "Akarsh"; integrated Chatterbox voice clone into an Infinite Talk ComfyUI workflow for end-to-end lip-synced video generation from a single image + text prompt; **~₹10–15 per 40s video** on Jarvis Labs GPU.
- JarvisLabs shipped an outdated ComfyUI template — manually installed the latest version via Jupyter, then exposed the server publicly using ngrok since it ran on a remote host with no direct URL.

02 AI Automation Agents (n8n) ↗ GitHub

n8n · OpenAI GPT-4.1 · Gmail API · Razorpay · Google Calendar · Sora2 · Google Sheets

- **WhatsApp Clinic Agent:** n8n webhook trigger → GPT-4.1 intent classification → Razorpay payment confirmation as conditional gate → Google Calendar booking; fully async, zero human input.
- **AI UGC Video Generator:** product photo-to-UGC-style video pipeline using Sora2 + GPT-4.1 mini at **$0.15/video**.
- **Auto Email Labeling Agent:** reads Gmail, classifies and labels emails, auto-drafts replies using GPT-4.1 nano.

03 Sahayak — AI Student Assistant (Hackathon) ↗ Backend ↗ Frontend ↗ Demo

FastAPI · Next.js · ChromaDB · Google Gemini Flash 2.5 · Google Vision API · GCP VM

- Built a full-stack RAG application that ingests textbooks via OCR, retrieves context through ChromaDB vector search, and uses Gemini Flash 2.5 to generate question papers and evaluate answer sheets.

04 Claude Code Agent Projects

Claude Code · Remotion · YouTube Data API · Google Sheets API · LLM APIs

- **YouTube Niche Analyzer:** agent pipeline that fetches video metadata, runs LLM analysis on engagement patterns, and auto-generates structured Excel + PPT reports delivered via email; configurable fetch count, reports in 5–10s. ↗ GitHub
- **Motion Graphics Generator:** prompt-driven pipeline using Claude Code + Remotion, rendering to MP4. ↗ GitHub
Google Sheet AI Chat: LLM-powered natural language Q&A over Sheets data. ↗ GitHub

EDUCATION

B.E. — Computer Science & Engineering
Nitte Meenakshi Institute of Technology, Bengaluru
CGPA: 8.16 / 10.0 · Jul 2020 – Jun 2024

CERTIFICATIONS

100xEngineers GenAI Program (In Progress)
Diffusion Models · LLM Pipelines · AI Agents · RAG